5.6.3 Condition for the line $y = mx + c$ to be a tangent to the conic sections

(i) parabola $y^2 = 4ax$

Let (x_1, y_1) be the point on the parabola $y^2 = 4ax$. Then $y_1^2 = 4ax_1$ ... (1)

Let $y = mx + c$ be the tangent to the parabola ... (2)

Equation of tangent at (x_1, y_1) to the parabola from 5.6.1 is $yy_1 = 2a(x + x_1)$. ... (3)

Since (2) and (3) represent the same line, coefficients are proportional.

$$\frac{y_1}{1} = \frac{2a}{m} = \frac{2ax_1}{c}$$

$$\Rightarrow y_1 = \frac{2a}{m}, x_1 = \frac{c}{m}$$

Then (1) becomes, $\left(\frac{2a}{m}\right)^2 = 4a\left(\frac{c}{m}\right)$

$$\Rightarrow \boxed{c = \frac{a}{m}}$$

So the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ and the equation of tangent to parabola is $y = mx + \frac{a}{m}$.

The condition for the line $y = mx + c$ to be tangent to the ellipse or hyperbola can be derived as follows in the same way as in the case of parabola.

(ii) ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Condition for line $y = mx + c$ to be the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$, with the point of contact is $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$ and the equation of tangent is $y = mx \pm \sqrt{a^2m^2 + b^2}$.

(iii) Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Condition for line $y = mx + c$ to be the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 - b^2$, with the point of contact is $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$ and the equation of tangent is $y = mx \pm \sqrt{a^2m^2 - b^2}$.

Note

(1) In $y = mx \pm \sqrt{a^2m^2 + b^2}$, either $y = mx + \sqrt{a^2m^2 + b^2}$ or $y = mx - \sqrt{a^2m^2 + b^2}$ is the equation to the tangent of ellipse but not both.

(2) In $y = mx \pm \sqrt{a^2m^2 - b^2}$, either $y = mx + \sqrt{a^2m^2 - b^2}$ or $y = mx - \sqrt{a^2m^2 - b^2}$ is the equation to the tangent of hyperbola but not both.

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